

PULKIT GARG

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AREA OF EXPERTISE

Machine Learning · Agentic AI · RAG Pipelines · LangChain · Prompt Engineering · API Integration · Data Pipelines

EDUCATION

Thapar Institute of Engineering and Technology, Patiala

Aug 2023 – May 2027

Bachelor of Engineering in Computer Science and Engineering

CGPA: 8.71

PROJECTS

SynthSales – Agentic AI Sales CRM

Tech: FastAPI, PostgreSQL, SQLAlchemy 2.0, Alembic, APScheduler, Redis, Docker, Google APIs, Gemini API

Agentic B2B outreach backend that turns uploaded company CSVs into researched, scored, contactable prospects with verified emails, AI-drafted outreach, reply tracking, and meeting scheduling.

- Built a FastAPI backend with **78** REST endpoints, **14** PostgreSQL models, JWT auth, admin approvals, and Alembic migrations.
- Engineered an **8-agent pipeline** for enrichment, scoring, contact discovery, email verification, outreach, tracking, meetings, and reply classification.
- Added safety systems including outbound kill-switches, do-not-contact suppression, **24h** undo snapshots, locked conversations, and Redis-backed rate limiting.

VOCA - Vocabulary Learning Platform

Tech: Next.js 14, MongoDB, React, LangChain, NextAuth.js, Tailwind CSS

Full-stack vocabulary learning platform for storing, organizing, and practicing English words with spaced repetition, progress tracking, and AI-assisted support.

- Constructed **14** authenticated API routes and **5** protected dashboard modules for word management, practice tracking, settings, and AI workflows.
- Designed a **6-level** spaced repetition system across **4** practice modes to power due reviews, streaks, daily goals, and mastery updates.
- Created a LangChain chatbot with persistent MongoDB memory, supporting **30** recent threads and replaying up to **50** messages per thread with one-click word saving.

Real-Time Self-Healing Data Pipeline

Tech: Apache Airflow, Python, PostgreSQL, dbt, Docker, Streamlit, Ollama LLaMA 3.2, pytest

Self-healing data pipeline that detects schema drift, repairs malformed CSV inputs with a local LLM, and monitors pipeline health.

- Developed a **4-task** Airflow pipeline for CSV generation, validation, PostgreSQL ingestion, and dbt transformation.
- Implemented LLaMA 3.2 self-healing to repair schema drift and log every AI intervention in PostgreSQL.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Backend & APIs: FastAPI, REST APIs, NextAuth.js

Machine Learning: Scikit-learn, TensorFlow, Regression & Classification, Neural Networks, NLP

Databases: MongoDB, MySQL, PostgreSQL, ChromaDB

AI & Automation: LangChain, LangGraph, LLM Integration, Ollama

Data Engineering: Apache Airflow, dbt, Streamlit

Tools & Version Control: Git, GitHub, Docker

CERTIFICATIONS & ACHIEVEMENTS

Supervised Machine Learning: Regression and Classification

Jan 2025

DeepLearning.AI & Stanford University (Coursera)

Advanced Learning Algorithms

Dec 2025

DeepLearning.AI & Stanford University (Coursera)